MAHIN FAREEHA

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RESEARCH PROFILE

Research-oriented Computer Science graduate passionate about advancing Artificial Intelligence through Computer Vision, Autonomous Systems, and Intelligent Robotics. My experience includes developing AI solutions for image understanding, multimodal learning, simulation environments, and intelligent automation, supported by published research and industry projects. I aspire to contribute to graduate research focused on embodied intelligence, autonomous perception, human-AI interaction, and scalable machine learning systems that bridge research innovation with real-world deployment.

RESEARCH INTERESTS

Artificial Intelligence · Autonomous Systems · Computer Vision · Deep Learning · Multimodal AI · Large Language Models (LLMs) · Intelligent Robotics · Machine Learning · Embodied AI

EDUCATION

Methodist College of Engineering & Technology, Hyderabad — B.E. Computer Science Engineering *May 2026*

CGPA: 8.29 / 10 | Final Year: 9.37

Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Software Engineering, Cryptography & Network Security, Cloud Computing, Digital Image Processing

Al Ain Juniors School, Al Ain, Abu Dhabi — High School *April 2021*

CGPA: 412/500 | Computer Science: 93/100

PUBLICATION

Haquani, M.F., "Automated 2D Blueprint-to-3D Gamified Tactical Training Simulator for Security Forces Using YOLOv8," International Journal for Research in Applied Science & Technology (IJRASET), Vol. 14, No. 4, April 2026.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, Java, C, SQL

AI / ML / Deep Learning: PyTorch, TensorFlow, scikit-learn, CNNs (VGG-16), LSTM, BERT, NLP, Sentiment Analysis, Random Forest, Model Evaluation & Cross-Validation, Anomaly Detection, Object Detection (YOLOv8)

LLMs & Agents: Large Language Models (LLMs), Prompt Engineering, LangChain, Retrieval-Augmented Generation, Multi-Agent Orchestration

Frontend: React.js, HTML5, CSS3, Tailwind CSS, Chart.js, Flutter (Dart)

Backend & APIs: Flask, FastAPI, Node.js, Express.js, RESTful API Design, JWT Authentication, Socket.IO

Data & Databases: Pandas, NumPy, Firebase, MySQL, Google Cloud Storage

DevOps & Cloud: Docker, Git & GitHub, Render, Vercel, Railway, Google Maps API

Simulation & Robotics-Adjacent: Unity, Procedural Generation, Custom Object Detection Datasets

EXPERIENCE

AI Engineering Intern — Evoastra Ventures Pvt. Ltd. *Jul 2026 – Oct 2026*

Python, TensorFlow, VGG-16, LSTM, BeautifulSoup

- Engineered an end-to-end multimodal image captioning system by integrating VGG-16 CNN for visual feature extraction with an LSTM-based sequence decoder, trained and evaluated on a curated 18.7K-image subset of the MS COCO dataset. Achieved competitive benchmark performance on the MS COCO dataset (CIDEr: 114, BLEU-4: 0.33), demonstrating strong semantic relevance and caption quality.
- Designed a scalable data preprocessing pipeline that cleaned and tokenized 93,950 image captions across a 12,819-word vocabulary, implementing feature extraction, normalization, sequence encoding, and serialized data storage for efficient training in memory-constrained cloud environments.
- Developed a BeautifulSoup-based web scraping pipeline to collect and analyze BBC News articles for a market intelligence case study, enabling automated trend monitoring and content analysis.

Software Engineering Intern — Codec Technologies Pvt. Ltd. *Sep 2024 - Nov 2024*

React.js, Node.js, Express.js, FastAPI, MySQL, MongoDB, REST APIs

- Led the end-to-end development of two production-ready AI-powered full-stack applications, independently driving architecture design, implementation, testing, and deployment.
- Built scalable web systems integrating machine learning models, real-time communication, secure authentication, and responsive user interfaces using modern full-stack technologies.
- Collaborated within an Agile development environment, applying software engineering best practices including modular architecture, API design, version control, and iterative delivery.
- Recognized with a Letter of Recommendation from the Program Manager and a Training Completion Certificate for outstanding technical performance and timely project completion.

RESEARCH & ACADEMIC PROJECTS

Secure3DOpsSim — Blueprint-to-3D Mission Simulator *Major Project · Research Publication*

Unity · Python · YOLOv8 Object Detection (Custom Blueprint Dataset) · Procedural Generation · Analytics · [github.com/RESCUESIM]

- Architected an FPS-style 3D simulation platform for security-force mission rehearsal, replacing costly physical drills with offline, site-specific digital training environments.
- Trained a custom YOLOv8 object detection model on a proprietary blueprint image dataset to automatically parse floor plans and classify entries, exits, staircases, and structural zones, substantially cutting manual scene-build time.
- Designed a mission-scripting engine with objective sequencing, spawn-point configuration, and trainee behavior logging; built post-mission analytics tracking path efficiency and decision latency per session.

VigilantEye — AI Telegram Surveillance & Monitoring System *Minor Project*

Python · Flask · BERT · Telegram API · JWT Auth · Google Cloud Storage · [github.com/VigilantEye-Drug-Detection-Bot-]

- Built an NLP monitoring pipeline with a fine-tuned BERT model for multi-class content classification, separated into ingestion, classification, and alerting service layers.
- Engineered a secure analyst dashboard with JWT-based auth, role-based access control, real-time alert surfacing, and GCS media archival for evidence-chain preservation.

AI Multi-Agent Orchestration System *Independent · Evaluated*

Python · LLM APIs · LangChain · Prompt Engineering

- Designed a task-decomposition framework using LLM APIs and prompt chaining; implemented inter-agent communication and retry logic, improving task completion reliability by ~30% over a single-agent baseline.

AI-Powered Expense Insight Dashboard *Codec Technologies · Internship*

React · FastAPI · scikit-learn · Random Forest · Chart.js · Tailwind CSS · [github.com/my-finance-app]

- Built a modular React SPA with a FastAPI backend performing Random Forest expense categorization (cross-validated) with anomaly detection and LLM-generated savings recommendations.
- Designed real-time Chart.js visualizations reducing time-to-insight from raw transaction data to under 3 seconds.

Full-Stack Real-Time Chat Application *Codec Technologies · Internship*

React · Node.js · Express · Socket.IO · JWT · OTP Auth · MySQL · [github.com/my-chat-app]

- Engineered a multi-room WebSocket platform sustaining 50+ concurrent connections with sub-200ms delivery, decomposed into auth, messaging, and notification service modules.
- Integrated AI-driven message suggestions, sentiment tagging, and automated moderation; implemented voice notes, file sharing, typing indicators, and read receipts.

CERTIFICATIONS

Microsoft: Azure AI Language (NLP), AI Careers for Women
IBM: AI Fundamentals, AI-Enabled Applications for Customer Service, Cybersecurity Fundamentals
Cisco Networking Academy: CCNA Networking Fundamentals, Python Essentials, IoT Fundamentals, Packet Tracer
GUVI (IIT Madras Research Park): Generative AI
Syscoms College: Data Science Certification (Industry Certification

LANGUAGES

English (Expert / C2) · Urdu (Native / C2) · Hindi (Advanced / B2) · Arabic (Basic / A2)